



3 Complex Analysis 2024

Tutorial 8

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Liouville's theorem

Problem 32 Let f be an entire function. Suppose that $|f(z)| \geq 1$ for all $z \in \mathbb{C}$. Show that f is constant.

Identity theorem

Problem 33

- (a) Is there a function f that is holomorphic at 0 and satisfies $f\left(\frac{1}{n}\right) = f\left(-\frac{1}{n}\right) = \frac{1}{n^2}$ for all $n \in \mathbb{N}$?
- (b) Is there a function f that is holomorphic at 0 and satisfies $f\left(\frac{1}{n}\right) = f\left(-\frac{1}{n}\right) = \frac{1}{n^3}$ for all $n \in \mathbb{N}$?

Gauss' mean value theorem

Problem 34 Prove Gauss' Mean Value Theorem:

Let f be a holomorphic function in the open disc $D(z; R)$ with some $z \in \mathbb{C}$, $R > 0$. Let $0 < r < R$. Then

$$f(z) = \frac{1}{2\pi} \int_0^{2\pi} f(z + re^{i\theta}) d\theta.$$

Problem 35 Evaluate $\int_0^{2\pi} \sin^2\left(\frac{\pi}{6} + 2e^{it}\right) dt$.

Maximum modulus principle

Problem 36

- (a) Let f be holomorphic in the disc $D(z_0; R)$ with some $z_0 \in \mathbb{C}$ and $R > 0$. If $|f|$ is constant in $D(z_0; R)$, then f is constant in $D(z_0; R)$. Hint: Use the Cauchy-Riemann Equations.

(b) Prove the Local Maximum Modulus Theorem:

Let f be holomorphic in the disc $D(z_0; R)$ with some $z_0 \in \mathbb{C}$ and $R > 0$. Assume that $|f(z)| \leq |f(z_0)|$ for all $z \in D(z_0; R)$. Then f is constant in $D(z_0; R)$.

(c) Prove the Maximum Modulus Principle:

Let G be a bounded region. Let f be holomorphic in G and continuous in $\overline{G}$. Then $|f|$ attains its maximum on the boundary of G which we denote by ∂G . If f is non-constant, then

$$|f(z)| < \max_{w \in \partial G} |f(w)| \quad \text{for any } z \in G.$$

Hint: Use the Local Maximum Modulus Theorem and the Identity Theorem.